Spectral decomposition for efficient smoothing-parameter selection in penalized least squares: an empirical study of complete-criterion computation

Heriberto Espino-Montelongo[1*]

[1*]Universidad de las Américas Puebla, San Andrés Cholula, Puebla, Mexico.

Corresponding author(s). E-mail(s): heriberto.espinomo@udlap.mx;

Abstract

Selecting the smoothing parameter of a finite-difference penalized least-squares estimator requires evaluation of a statistical selection criterion, not merely solution of a penalized linear system. We examine the computational advantages and numerical limitations of reusing a singular-value decomposition (SVD) across candidate parameters, compared with a structured banded-Cholesky implementation. The study covers 29 prespecified configurations, 97 random-seed realizations per configuration, multiple selection criteria and one- versus multiple-candidate workloads. Valid execution times are matched by realization, workload and execution phase, with uncertainty estimated by seed-cluster bootstrap. For an order-three conditional cross-validation problem with $\boldsymbol{n = 256}$, the banded/SVD complete cold-time ratio is 0.698 (95% interval 0.671–0.724) for a single candidate but 4.846 (4.567–5.125) for a 100-candidate grid. In nine configurations, the banded complete-grid implementation fails its post-solve optimality check for all recorded grid evaluations while the spectral pipeline completes them; both methods exhibit smoothness-inversion failures in larger fourth-order configurations. Dense spectral storage nevertheless scales quadratically with series length. The results support spectral decomposition as the preferred implementation for repeated complete-criterion searches when memory permits, while retaining banded methods for isolated evaluations and memory-constrained problems.

Keywords: penalized least squares, smoothing-parameter selection, singular-value decomposition, Whittaker–Henderson smoothing, cross-validation, numerical reliability, computational performance

1 Introduction

Penalized least-squares smoothing is a widely used approach to separating low-frequency variation from noise in an observed series. Finite-difference penalties have connected trend regularity and linear smoothing since early work on statistical graduation (Whittaker 1923). Subsequent developments include structured implementations of Whittaker–Henderson smoothing (Weinert 2007, 2012; Biessy 2026) and the interpretation of smoothing through a normalized percentage-of-smoothness parameter (Guerrero 2007, 2008). Selection of the penalty by cross-validation, information criteria or marginal likelihood is well established (Craven and Wahba 1979; Cortés-Toto et al. 2017). Nevertheless, obtaining a fitted trend and evaluating the *complete selection criterion* are computationally different tasks.

A banded Cholesky factorization exploits the sparsity of finite-difference penalties and is economical for an individual smoothing parameter. In contrast, a spectral factorization requires an appreciable initial calculation but exposes all candidate penalties through diagonal transformations in a common basis. Consequently, the apparent superiority of a solver in a single fit need not persist across a grid of smoothing parameters. Moreover, numerical acceptance of the complete criterion depends not only on the matrix factorization but also on leverage calculations, endpoint treatment and the inversion between penalty and smoothness coordinates.

We investigate the following practical question: *When is a spectral implementation preferable for selecting a smoothing parameter by a complete statistical criterion?* We compare measured end-to-end costs of spectral singular-value decomposition (SVD), banded Cholesky and available reference implementations under paired computational workloads. Unlike a fit-only benchmark, the comparisons charge setup, coordinate conversion, criterion evaluation and diagnostics to the method that incurs them. Unsuccessful or incomplete searches remain observable outcomes rather than being treated as fast computations. This permits a recommendation based jointly on execution time, observed reliability and storage requirements.

Our main conclusion is that spectral decomposition is an effective *default choice for repeated, complete-criterion searches at feasible matrix sizes.* That recommendation is empirical and conditional: structured banded computation remains competitive for isolated candidate evaluations and retains important storage advantages for long series. We do not propose a new PLS estimator, nor do we interpret computation time as predictive accuracy.

2 Penalized least squares and spectral evaluation

2.1 Estimator and smoothing coordinate

Let $y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ denote equally spaced observations and let $D_d \in \mathbb{R}^{(n-d)\times n}$ be the order-d forward-difference matrix, with $1 \le d < n$. For a fixed reference level $\mu \in \mathbb{R}$, the trend estimate is

$$\widehat{\tau}_{\lambda,\mu} = \underset{\tau\in\mathbb{R}^n}{\arg\min} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0. \tag{1}$$

Writing $Q = D_d^\top D_d$ and choosing any p_μ satisfying $D_d p_\mu = \mu \mathbf{1}$, the normal equations yield

$$\widehat{\tau}_{\lambda,\mu} = p_\mu + H_\lambda(y - p_\mu), \qquad H_\lambda = (I + \lambda Q)^{-1}. \tag{2}$$

For finite $\lambda \geq 0$, $I + \lambda Q$ is symmetric positive definite. Because $\operatorname{rank}(D_d) = n - d$, the penalty matrix has precisely d zero eigenvalues. These correspond to discrete polynomials of degree at most $d - 1$.

Let $\delta_1, \ldots, \delta_{n-d} > 0$ be the positive eigenvalues of Q. The normalized smoothness parameter

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n - d} = \frac{1}{n - d} \sum_{j=1}^{n-d} \frac{\lambda \delta_j}{1 + \lambda \delta_j} \tag{3}$$

satisfies $S(0) = 0$ and $S(\infty) = 1$, and is strictly increasing for positive finite λ. Thus S supplies a bounded coordinate for the original estimator, without changing the underlying objective function. The endpoint $S = 1$ is handled as the constrained polynomial limit rather than by substituting an arbitrarily large finite penalty.

2.2 One decomposition, multiple smoothing parameters

Consider the singular-value decomposition $D_d = U \Sigma V_+^\top$, completed to an orthogonal basis $V = [V_+, V_0]$. Its nonzero squared singular values are the δ_j. With $a = V_+^\top (y - p_\mu)$ and $w_j(\lambda) = \lambda \delta_j / (1 + \lambda \delta_j)$, the fitted trend and residual can be represented as

$$\widehat{\tau}_{\lambda,\mu} = y - V_+ \operatorname{diag}\{w_j(\lambda)\} a, \tag{4}$$

$$r_\lambda = y - \widehat{\tau}_{\lambda,\mu} = V_+ \operatorname{diag}\{w_j(\lambda)\} a. \tag{5}$$

After the basis and series-specific coordinates have been computed, each candidate modifies only the scalar attenuation factors. The effective degrees of freedom and residual sum of squares follow from

$$\operatorname{edf}(\lambda) = d + \sum_{j=1}^{n-d} \{1 - w_j(\lambda)\}, \tag{6}$$

$$\operatorname{RSS}(\lambda) = \sum_{j=1}^{n-d} w_j(\lambda)^2 a_j^2. \tag{7}$$

In particular, a criterion requiring only RSS and edf can exploit the stored spectral coordinates without repeatedly factorizing a penalized normal matrix. This amortization is the central reason to consider SVD for a grid search.

2.3 Complete selection criteria

For a fixed numerical reference mean, ordinary leave-one-out cross-validation (CV) and generalized cross-validation (GCV) have the conventional forms

$$\operatorname{CV}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{r_{\lambda,i}}{1 - h_{ii}(\lambda)} \right)^2, \tag{8}$$

$$\mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}(\lambda)/n}{\{1 - \mathrm{edf}(\lambda)/n\}^2}, \tag{9}$$

where $h_{ii}(\lambda)$ is a diagonal entry of H_λ. The spectral basis also gives the stable leverage-deficit representation

$$1 - h_{ii}(\lambda) = \sum_{j=1}^{n-d} v_{ij}^2 w_j(\lambda). \tag{10}$$

Unlike GCV, ordinary CV requires observationwise residuals and leverages; consequently, the candidate-evaluation cost differs by criterion even after spectral setup. We additionally consider corrected Akaike information criterion (AICc), Bayesian information criterion (BIC), and profiled Gaussian marginal likelihood. These use the established definitions implemented consistently across the methods being compared, including their prescribed undefined points and endpoint limits. A criterion value of infinity is not interpreted as an executable failure when the criterion is mathematically undefined.

A reference mean estimated once as $\mu = \overline{D_d y}$ is used in some computational settings. In those cases the selection criteria are *conditional* on the full-sample plug-in value: the CV score does not re-estimate μ within each leave-one-out fold. This distinction is preserved throughout the analysis.

2.4 Banded computation and verification of numerical optimality

The banded implementation factorizes $A_\lambda = I + \lambda D_d^\top D_d$ for individual finite positive penalties. A selected-inverse recursion recovers the diagonal leverage values and effective degrees of freedom needed by the complete criteria, without assembling a dense inverse. At fixed low difference order, the banded factor and selected inverse require $O(nd)$ storage, compared with $O(n^2)$ for a dense spectral basis. A dense explicit-inverse reference and a sparse-factorization/structured diagnostics hybrid are included as additional computational comparisons; the latter does not constitute an entirely sparse criterion evaluator.

Optimality equation.

For the strictly convex and unconstrained objective in Eq. (1), the first-order condition is both necessary and sufficient for global optimality (Boyd and Vandenberghe 2004):

$$\begin{aligned} \frac{1}{2}\nabla J_\lambda(\tau) &= (\tau - y) + \lambda D_d^\top (D_d \tau - \mu \mathbf{1}) \\ &= A_\lambda \tau - b_\lambda = 0, \\ b_\lambda &= y + \lambda \mu D_d^\top \mathbf{1}. \end{aligned} \tag{11}$$

In this setting, the expression often called the Karush–Kuhn–Tucker (KKT) condition reduces simply to *stationarity*: there are no explicit feasibility, complementarity, or dual-multiplier conditions. The Hessian is $2A_\lambda \succ 0$, and thus every finite $\lambda \geq 0$ possesses a unique minimizer τ_λ^*.

Post-solve optimality residual.

Given a computed solution $\widetilde{\tau}_\lambda$, define

$$\begin{aligned} r_{\text{opt}} &= b_\lambda - A_\lambda \widetilde{\tau}_\lambda \\ &= (y - \widetilde{\tau}_\lambda) - \lambda D_d^\top (D_d \widetilde{\tau}_\lambda - \mu \mathbf{1}) \\ &= -\tfrac{1}{2} \nabla J_\lambda(\widetilde{\tau}_\lambda). \end{aligned} \tag{12}$$

Assessing a computed solution through the residual of its optimality or linear-system equations is established numerical practice, as developed in classical treatments of linear-system residuals and backward errors (Higham 2002; Golub and Van Loan 2013). Residual-based stopping or acceptance rules are also used in practical convex quadratic-programming solvers (Stellato et al. 2020). These references justify the *type of diagnostic*, not the particular numerical tolerance selected for the present implementation.

Specifically, after a successful `cholesky_banded` factorization and `cho_solve_banded` solve, the evaluated banded procedure computes the following *scaled first-order optimality residual*:

$$\boxed{\eta_{\text{opt}} = \frac{\|r_{\text{opt}}\|_2}{\max\{1, \|y\|_2, \|b_\lambda\|_2\}}.} \tag{13}$$

A finite-penalty criterion evaluation is rejected if the computed trend or diagnostic is nonfinite, or if $\eta_{\text{opt}} > 10^{-7}$. The threshold is an *implementation-specific acceptance rule*; it is not presented as a canonical KKT tolerance prescribed by the cited numerical-analysis literature. The zero-penalty and infinite-penalty endpoints are handled by separate analytic branches and do not pass through this finite-penalty residual check.

Distinction from backward and forward errors.

Although the numerical code labels this diagnostic a `backward_error`, its normalization is *not* the conventional normwise linear-system backward error, which can instead be expressed, for the present norms, as shown in classical backward-error analysis (Higham 2002):

$$\eta_{\text{lin}} = \frac{\|b_\lambda - A_\lambda \widetilde{\tau}_\lambda\|_2}{\|A_\lambda\|_2 \|\widetilde{\tau}_\lambda\|_2 + \|b_\lambda\|_2} \tag{14}$$

The distinction matters: the recorded threshold is a reproducible residual-based *acceptance convention*, not a formal bound on relative forward error or a certified backward-stability statement.

For this quadratic objective, however, the mathematical meaning of the *unscaled* optimality residual is exact. With $e = \widetilde{\tau}_\lambda - \tau_\lambda^*$, Eq. (12) gives $e = -A_\lambda^{-1} r_{\text{opt}}$ and

$$\begin{aligned} J_\lambda(\widetilde{\tau}_\lambda) - J_\lambda(\tau_\lambda^*) &= e^\top A_\lambda e \\ &= r_{\text{opt}}^\top A_\lambda^{-1} r_{\text{opt}} \le \|r_{\text{opt}}\|_2^2. \end{aligned} \tag{15}$$

The inequality follows from $\lambda_{\min}(A_\lambda) = 1$; likewise $\|e\|_2 \le \|r_{\mathrm{opt}}\|_2$ in exact arithmetic. Neither inequality turns the implemented *scaled* criterion into a relative-accuracy guarantee, and floating-point evaluation of the residual itself incurs rounding error.

Validation of a complete criterion search.

For each requested finite penalty, the banded procedure (i) factorizes A_λ and solves for the trend, (ii) checks the scaled optimality residual, (iii) calculates selected-inverse entries, effective degrees of freedom and observationwise leverage quantities when required, and (iv) evaluates the statistical selection criterion. Conversion from a requested smoothness S to λ is validated separately; inaccurate inversion does not count as a KKT rejection. Completion of a full search requires all prescribed candidate–series evaluations and a finite, admissible selected score. Failure at one required candidate cannot be replaced by the runtime of a shorter, successful subset. Numerical checks belong to the overall computational workload rather than being excluded from the measured elapsed time.

Finally, the spectral implementation does not execute the *identical* banded-specific optimality-residual acceptance test. The observed contrast in successful grid completion therefore measures practical pipeline admissibility, not a controlled proof that the SVD solutions have smaller forward error. Independent high-precision or identical-diagnostic comparisons would be needed to establish such an accuracy ranking.

3 Computational study

3.1 Experimental factors and execution environment

The study evaluates 29 prespecified computational configurations and 97 independent random-seed realizations per configuration, giving 2813 scheduled case–seed jobs. Configurations vary series length n, difference order d, criterion, reference mean, signal amplitude, and grid density M. Four workload types are considered: one candidate for one series, M candidates for one series, one candidate for R series, and M candidates for R series. The tested grid is $S_j = j/M$ for $j = 1, \ldots, M$, including its analytic endpoint. Selected configurations use $M = 100$, 1000, or 10000; the examined sample lengths include $n = 64$, 128, 256, 512, 1024, 2048, and 4096. Comparisons use identical simulated series and evaluation coordinates when both implementations are admissible.

The computations were performed on a Windows x64 multicore workstation, with one BLAS thread per job. The execution schedule comprised phases allowing at most 24, 1, or 12 concurrent jobs, with 2231, 52, and 530 jobs, respectively. These limits do not measure instantaneous CPU occupancy. We therefore retain the execution phase as a matching variable and avoid interpreting multi-job wall-clock times as contention-free single-process latency. Within a case–seed, method ordering was rotated and repeated where prescribed.

Two coordinate-evaluation policies were implemented: directly inverting S for each requested candidate and converting the full S grid to λ once for reuse. Both policies target the *same* smoothing values. The reported complete cold time includes method

initialization, smoothness-to-penalty conversion and all candidate criterion evaluations; recorded warm accounting excludes the one-time initialization term but does not represent a separately launched process.

3.2 Complete-search validity and paired comparisons

Every scheduled job was reconciled with its saved numerical and timing outcomes. A timing row is admitted only when the process terminates successfully, the selected score and elapsed time are finite, the recorded status is successful, and all MR prescribed evaluations are present. Missing files, rejected numerical checks, inaccurate smoothness inversions and resource-limit exclusions remain in the denominator of descriptive reliability summaries. Successful evaluation of only one grid point is never substituted for completion of the full grid.

For paired comparisons, methods must additionally agree on selected smoothness within 10^{-10} and, for directly comparable non-likelihood scores, on the selected criterion value within relative tolerance 10^{-7}. These criteria assess agreement between available double-precision implementations, not uniform high-precision accuracy throughout the parameter grid. The implementation and machine configuration, execution phase, realization, coordinate policy, workload and repeat are matched.

Let T_{B} and T_{S} denote positive complete cold times for banded Cholesky and spectral SVD. Our primary contrast is

$$\rho = \frac{T_{\mathrm{B}}}{T_{\mathrm{S}}}; \qquad \rho > 1 \iff \text{SVD is faster for that paired workload.} \tag{16}$$

Log ratios are averaged within each independent seed cluster before constructing the geometric mean. Two-sided percentile 95% confidence intervals use 700 bootstrap resamples of the independent seeds, retaining the correlated within-seed repetitions together. Confidence intervals thus characterize run-to-run variation for the measured conditions, not performance uncertainty across hardware platforms. Conditioning on numerically successful pairs can introduce survivor bias, which is addressed by reporting reliability separately from speed.

4 Empirical results

4.1 Completion rates and admissible observations

Of 2813 scheduled computational jobs, 57 terminated without a successful numerical record. The resulting dataset comprises 92,679 measured or explicitly classified method-level outcomes. A total of 76,220 (82.24%) satisfy the predefined completeness and status conditions. The remaining 16,459 comprise 11,208 numerical-rejection records, 4224 additional smoothness-inversion failure records, 970 computational-budget exclusions, and 57 records representing unavailable job inputs. The classification retains unsuccessful searches; omitting them would bias subsequent comparisons toward implementations with a higher apparent completion rate.

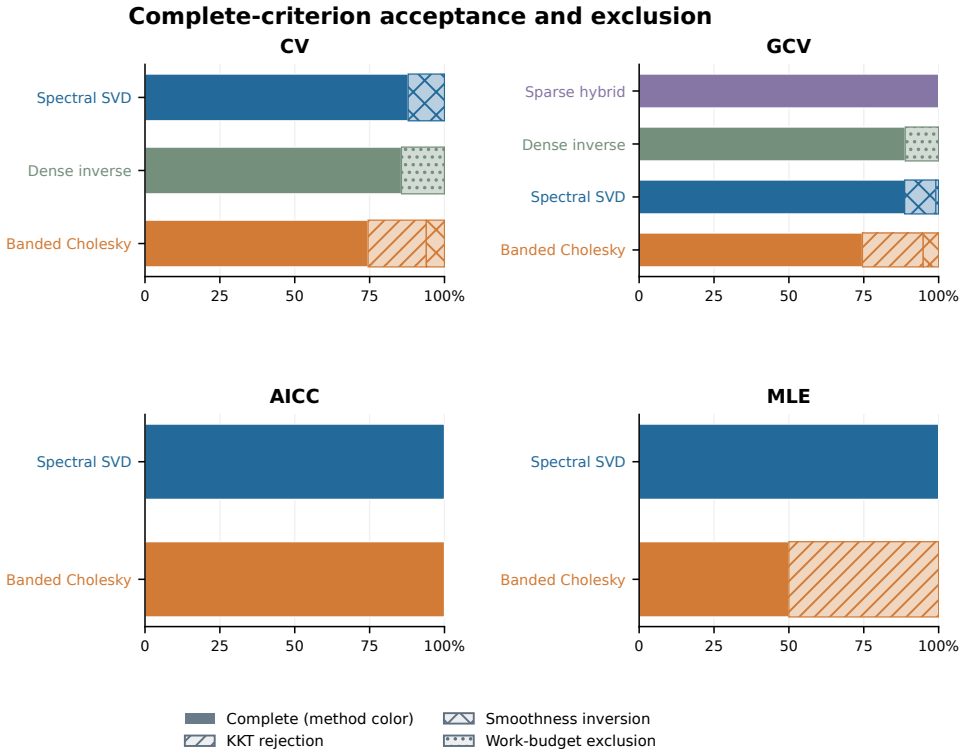


Fig. 1 Stacked status composition of complete-criterion evaluations by selection criterion. Solid colors identify the implementation, while patterned segments distinguish the recorded failure mechanisms. Rates are conditional on method-specific eligible observations and do not constitute matched failure probabilities.

Figure 1 displays stacked status fractions by criterion and implementation, with distinct colors identifying the methods and patterned segments indicating numerical rejection, inversion failure, or resource exclusion. Table 1 reports the underlying counts by method. Because eligibility varies across configurations, these percentages summarize observed method-level records rather than randomized or matched failure probabilities.

4.2 Complete-grid running time

Table 2 summarizes the matched cold-time ratios for eight complete 100-candidate CV and GCV searches, reporting both seed-level median and interquartile range and the previously calculated geometric mean with seed-cluster bootstrap intervals. Figure 2 displays the corresponding distributions as violin densities, narrow boxplots and individual seed-level paired ratios. Each case uses identical observations and candidates for both methods. Ratios $\rho = T_B/T_S > 1$ favor spectral SVD; all eight representative grids show this advantage. The distributions reveal variation and asymmetry hidden

Table 1 Recorded outcome classifications by numerical method, including all execution phases. Counts refer to correlated method-level observations rather than independent numerical trials. An additional 57 unavailable jobs cannot be assigned to a method.

Method	Recorded	Complete (rate)	KKT	S inversion	Work gate
Banded Cholesky	42,544	31,336 (73.7%)	9,096	2,112	0
Spectral SVD	42,350	37,932 (89.6%)	0	4,224	194
Dense inverse	6,176	5,400 (87.4%)	0	0	776
Sparse hybrid	1,552	1,552 (100.0%)	0	0	0

Table 2 Valid, matched 100-candidate one-series complete cold-time ratios $T_{\mathrm{B}}/T_{\mathrm{S}}$ at 24-worker phase. Medians and interquartile ranges describe the distribution of per-seed ratios; 95% intervals are seed-cluster bootstrap intervals for the geometric means.

Criterion	(n, d)	μ	Seeds	Median [IQR]	Geom. mean [95% CI]
CV	64, 1	0	96	2.37 [2.25, 2.46]	2.34 [2.29, 2.39]
CV	64, 3	0	97	4.07 [3.95, 4.27]	4.03 [3.95, 4.11]
CV	128, 3	0	97	5.56 [5.12, 5.81]	5.40 [5.24, 5.55]
CV	128, 3	$\overline{D_d y}$	97	5.59 [5.40, 5.74]	5.56 [5.46, 5.65]
CV	256, 3	$\overline{D_d y}$	97	5.52 [3.82, 5.92]	4.85 [4.57, 5.12]
GCV	64, 1	0	96	2.36 [2.26, 2.45]	2.33 [2.28, 2.37]
GCV	128, 3	0	97	5.12 [4.74, 5.34]	4.99 [4.87, 5.10]
GCV	128, 3	$\overline{D_d y}$	97	5.15 [5.01, 5.32]	5.12 [5.01, 5.23]

by the point summaries and must not be interpreted as eight identically distributed experiments.

The effect of candidate count is particularly clear for conditional CV at $n = 256$ and $d = 3$. The banded/SVD time ratio is 0.698 (95% interval 0.671–0.724) for one candidate and one series, but 4.846 (4.567–5.125) for 100 candidates. For 100 candidates evaluated on three series, the corresponding ratio is 3.655 (3.450–3.860). The reversal is consistent with a sizable spectral setup cost that is recovered through repeated criterion evaluation. It does not imply a universal crossover at $M = 100$; both the location and even the existence of a crossover depend on the criterion, series length, order and implementation.

For fixed-mean first-order GCV at $n = 64$, the analogous ratio is 1.001 (0.982–1.018) for one candidate and 2.328 (2.282–2.374) for the 100-candidate grid. Thus a negligible single-candidate performance difference can coexist with a substantial grid-search advantage.

4.3 Reuse across smoothing coordinates and series

The candidate representation also influences total cost. Two computational policies were compared: repeated conversion of smoothness values S to the penalty λ, and a single conversion shared across the search. The candidate set is identical under

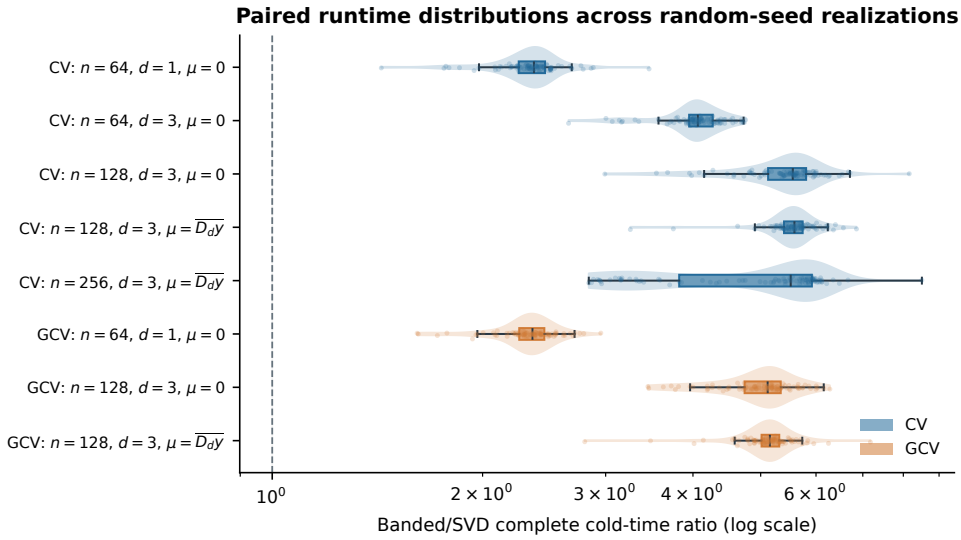


Fig. 2 Observed distributions of matched banded-Cholesky/SVD complete cold-time ratios over 100 smoothing candidates. The violin outlines are descriptive kernel-density estimates, the boxes give medians and interquartile ranges, and the dots represent independent-seed cluster summaries. The dashed line denotes equal time; numerical confidence intervals for geometric means are given in Table 2.

both policies, so the contrast concerns computational reuse rather than a different optimization problem.

For spectral CV with $n = 256$, $d = 3$, 100 candidates and three series, the geometric complete cold-time ratio of once-mapped penalties to repeated direct-S conversion is 0.580 (95% interval 0.570–0.590). In this setting, precomputing the candidate transformation reduces total execution time. Figure 3 displays corresponding descriptive comparisons across other numerically admissible multi-series grid configurations.

Figure 4 illustrates how a fixed first-order GCV configuration responds to four combinations of candidate and series count. The comparisons show why initialization, single-candidate scoring, multiple-series reuse and full-grid selection must be distinguished when assessing an implementation. In particular, a procedure optimized only for one fitted trend need not minimize the cost of selecting a smoothing parameter.

4.4 Numerical reliability and its consequences

The aggregate status proportions in the 24-worker execution phase are 10,848/13,176 (82.3%) for banded CV and 13,176/13,176 (100%) for spectral CV. The corresponding GCV counts are 13,176/16,280 (80.9%) and 15,892/16,086 (98.8%), respectively. Their denominators differ because method eligibility and the evaluated configurations differ; they should not be used as an unconditional estimate of solver accuracy. The comparison nevertheless motivates a closer examination of the failure mechanisms, reported next.

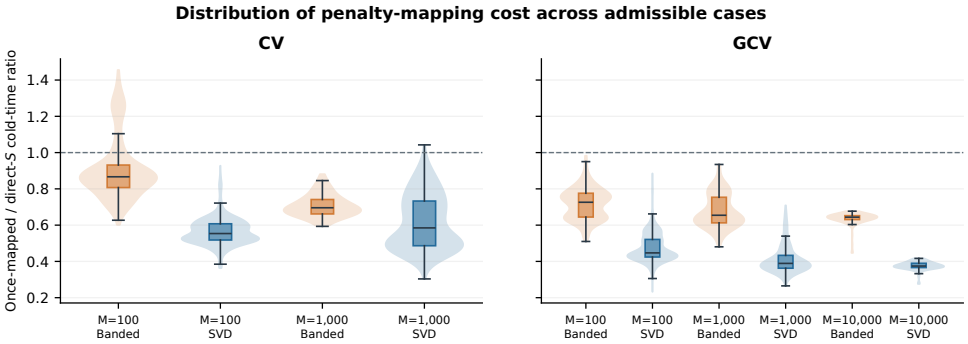


Fig. 3 Per-case-seed distributions of mapped-penalty/direct-smoothness cold-time ratios, grouped by criterion, method and candidate-grid size. Violin density and boxplots describe heterogeneous admissible cases under the common execution phase. Values below one favor mapping the penalty grid once; the pooled descriptive distributions are not universal algorithmic scaling laws.

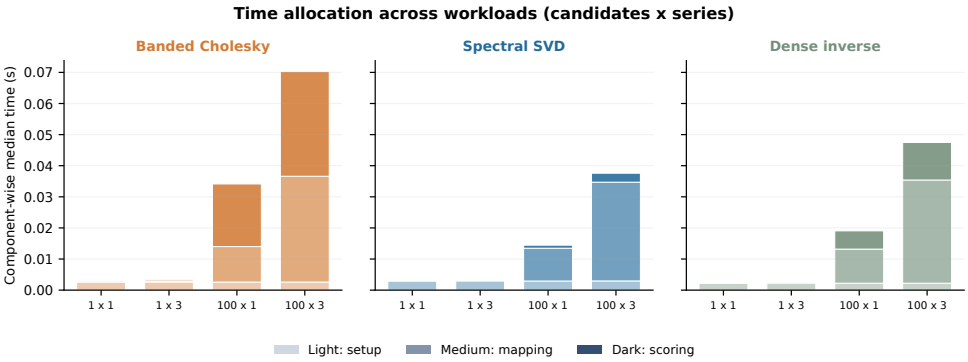


Fig. 4 Stacked component-wise medians for initialization, smoothness-to-penalty mapping and criterion evaluation in a first-order GCV example ($n = 64$, $M = 100$). Colors identify the implementation; light, medium and dark sections identify stages. Workload labels denote number of smoothing candidates times number of series. Sums of component medians are not generally the medians of total cold time.

4.5 Post-solve optimality failures in banded computation

The banded-Cholesky implementation recorded 9096 method-level timing observations rejected by the scaled stationarity-residual check in Eq. (13). They represent 3384 observations under CV, 4160 under GCV, and 1552 under marginal likelihood. Among the *rejected* records, the smallest reported value of η_{opt} was 1.0013×10^{-7}, the median was 1.4079×10^{-7}, and the largest was 1.0587×10^{-6}, against the specified threshold 10^{-7}. Figure 5 displays this conditional distribution. The data do not record the corresponding residual magnitudes for accepted runs; the histogram must not be used as an estimate of the full residual distribution or of an independent failure probability.

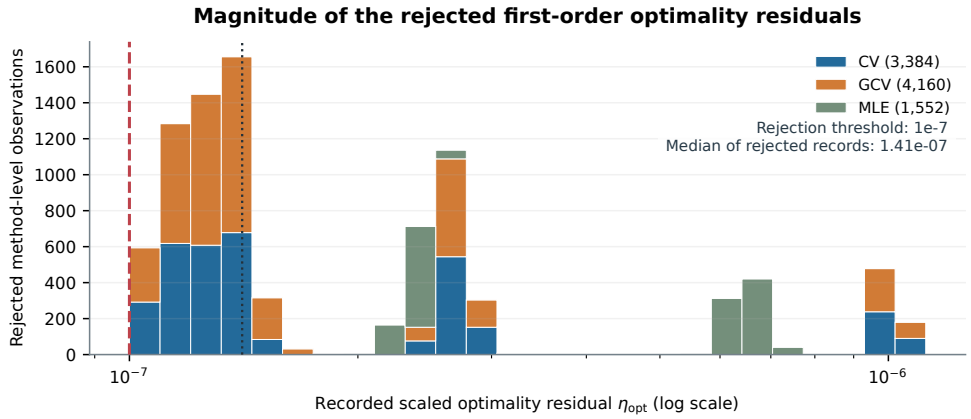


Fig. 5 Distribution of the scaled first-order optimality residual η_{opt} among *rejected* banded complete-criterion computations. Histogram segments are colored by statistical criterion (CV, GCV and Gaussian marginal likelihood). The dashed line marks the implementation-specific 10^{-7} acceptance threshold; the dotted line marks the median of rejected records. Counts are correlated method-level timing observations, not independent failures. Residuals of accepted evaluations are not available in the archived record.

Since the check follows factorization and solution, the failures establish rejection by the *complete banded criterion pipeline*, not an inability of Cholesky to factor an SPD matrix.

A particularly informative contrast occurs in nine parameter configurations. For each configuration, 97 independent realizations were considered. Both methods completed every recorded single-candidate evaluation, but the banded implementation rejected every complete-grid evaluation, whereas the spectral implementation completed the corresponding grids. The configurations comprise CV and GCV with $(n, d, M) = (128, 3, 1000)$, $(256, 4, 100)$ and $(256, 3, 1000)$; GCV with $(64, 3, 10000)$; and profiled Gaussian marginal likelihood with $(256, 3, 100)$ and $(512, 3, 100)$. Figure 6 displays these results.

The nine configurations account for 6984 banded-rejection timing records; another 2112 such records arise from larger fourth-order configurations. These are repeated method-level observations within 97 independent seed clusters per configuration, rather than thousands of independent experimental realizations. The saved status information does not identify the exact smoothing-grid point that first produced each rejection, and hence no candidate-specific instability boundary is inferred.

The apparent spectral advantage must be interpreted as *observed computational completion*, not a controlled comparison of identical backward-error tests. The spectral scorer does not impose the same implementation-specific acceptance criterion as the banded scorer. Independent numerical reference calculations would be required to establish a strict relative-accuracy ranking at each penalty.

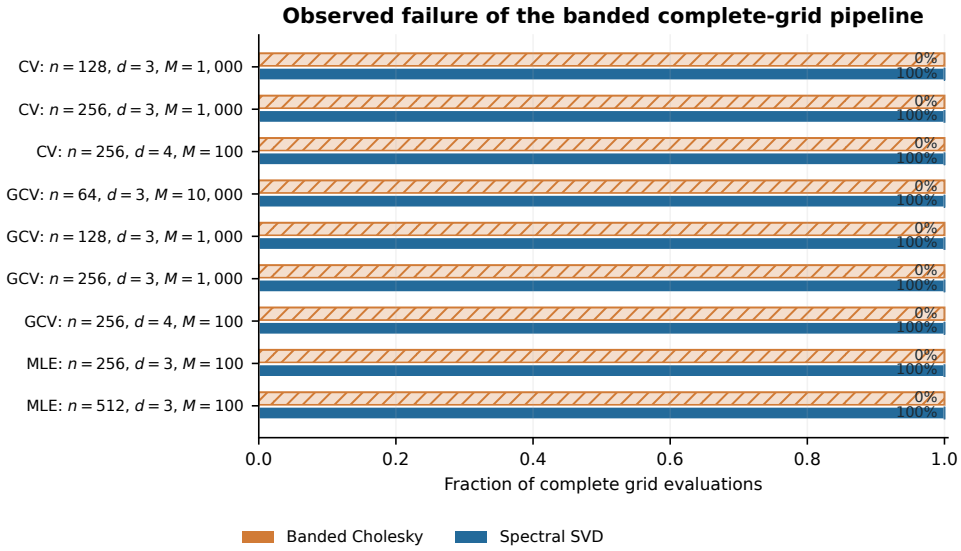


Fig. 6 Full-grid numerical completion in nine parameter configurations, with method-specific coloring and patterned rejection fractions. All recorded single-candidate outcomes complete for both implementations, but the banded full-grid pipeline is rejected while the spectral pipeline completes. Each configuration has 97 independent seed clusters; within-seed repeats, mapping policies and multiple-series workloads are correlated.

4.6 Smoothness inversion remains a common numerical limitation

An additional failure mechanism affects both implementations at larger difference order and series length. In the fourth-order CV/GCV configurations with $n = 1024$, 2048 and 4096, the numerical inversion of the normalized smoothness mapping sometimes returned values outside its prescribed tolerance. Across the evaluated configurations, 6336 timing records explicitly describe inaccurate smoothness inversion: 2112 associated with the banded method and 4224 with SVD. The distribution of observed outcomes in the fourth-order settings is given in Fig. 7.

This error concerns the transformation from the bounded coordinate S to the penalty λ, rather than evidence that a spectral decomposition failed. Crucially, a successfully completed one-candidate evaluation cannot be used as a substitute for an incomplete multi-candidate search. Accordingly, the large-order configurations with shared inversion difficulties do not supply an empirical winner for the full criterion grid.

4.7 The single-candidate versus grid-search crossover

The principal runtime trade-off becomes visible when paired, numerically admissible observations are compared across candidate counts. Figure 8 shows three examples evaluated under identical matching rules. In conditional CV with $n = 256$, $d = 3$, the

Table 3 Full-grid completion in nine configurations showing a post-solve banded KKT rejection. Each method has 776 correlated method-level grid observations (97 independent seeds, two repeats, two coordinate policies and two series workloads). Both methods completed all single-candidate observations.

Criterion	n	d	M	Banded complete	SVD complete
CV	128	3	1,000	0/776	776/776
CV	256	3	1,000	0/776	776/776
CV	256	4	100	0/776	776/776
GCV	64	3	10,000	0/776	776/776
GCV	128	3	1,000	0/776	776/776
GCV	256	3	1,000	0/776	776/776
GCV	256	4	100	0/776	776/776
MLE	256	3	100	0/776	776/776
MLE	512	3	100	0/776	776/776

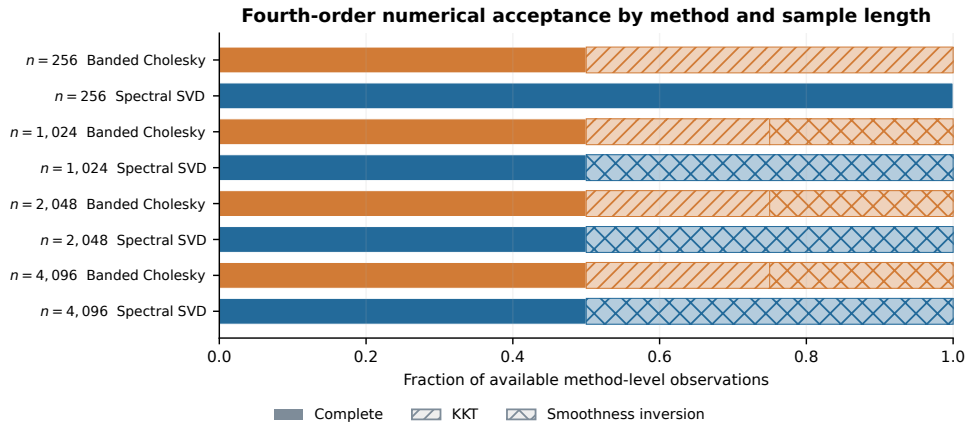


Fig. 7 Stacked method-colored fractions of numerically complete evaluations and distinct rejection mechanisms for difference order four. The bands show implementation-specific complete, post-solve KKT-rejected and inaccurate smoothness-inversion outcomes. Denominators consist of available method-level records; both implementations exhibit inversion failures at higher sample lengths.

geometric banded/SVD cold-time ratio changes from 0.698 (95% interval 0.671–0.724) for one candidate to 4.846 (4.567–5.125) for 100 candidates. In fixed-mean GCV with $n = 64$, $d = 1$, the corresponding ratios are 1.001 (0.982–1.018) and 2.328 (2.282–2.374). For fixed-mean CV with $n = 128$, $d = 3$, they are 1.069 (1.052–1.087) and 5.401 (5.244–5.555).

These results support the computational interpretation of Eqs. (4)–(7): spectral setup is paid once, while the attenuation coefficients and the statistics needed for subsequent criterion evaluations are reused. The crossover is observed in complete scoring workloads; it should not be conflated with the cost of fitting a trend for a prescribed penalty.

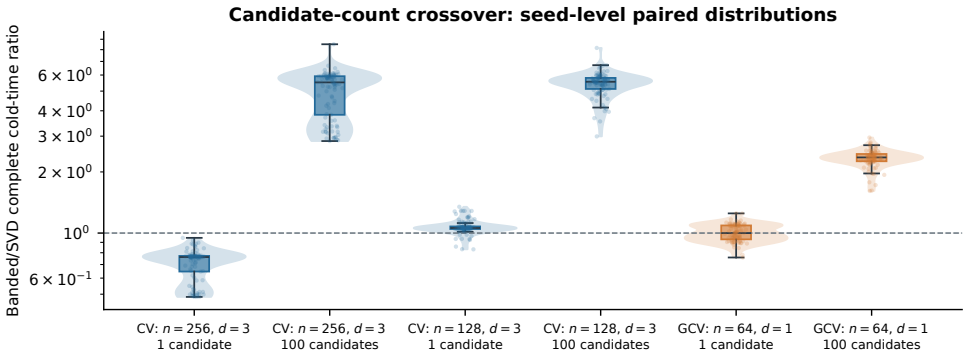


Fig. 8 Distributional comparison of matched banded-Cholesky/SVD cold-time ratios for one candidate and 100 candidates. Violin densities, boxplots and individual seed-level points show the pronounced reversal of computational preference. Values above one favor SVD. Corresponding independent-seed bootstrap intervals for geometric means are reported in the text rather than inferred from violin widths.

4.8 Memory requirements and computational feasibility

The principal disadvantage of the spectral approach is its dense storage requirement. For the evaluated implementation, the conservative memory admission allowance corresponds to five $n \times n$ double-precision arrays, or $40n^2$ bytes. An explicit dense-inverse reference is assigned seven such arrays, or $56n^2$ bytes. In contrast, a banded Cholesky factor and a banded selected-inverse buffer together require approximately $16n(d+1)$ bytes, excluding ancillary data structures. Figure 9 illustrates these storage relations for $d = 4$.

For example, the spectral admission allowance is 640 MiB at $n = 4096$. At $n = 8192$ it rises to 2.5 GiB, exceeding the 2 GiB dense allocation limit imposed in the study. The latter sample length was not evaluated in the present comparison. The curves therefore represent *implementation-specific memory allowances and illustrative buffer sizes*, not measurements of peak resident memory or empirical allocation failures.

There were 970 resource-budget exclusions in the recorded timing outcomes: 776 for the explicit-inverse reference and 194 for SVD. These were induced by prescribed computational-work thresholds, not observed out-of-memory errors. Thus the memory analysis establishes a concrete feasibility trade-off without attributing numerical failures to unsupported memory measurements.

5 Discussion

The empirical evidence supports a clear implementation preference, subject to mathematical and computational feasibility: *reuse of a spectral decomposition is the recommended default for finite-difference PLS smoothing-parameter selection over a moderate-to-large candidate grid.* In contrast to a sequence of penalized normal-equation factorizations, spectral preprocessing represents the complete search in one orthogonal basis. It amortizes setup, supplies degrees of freedom and residual sums of

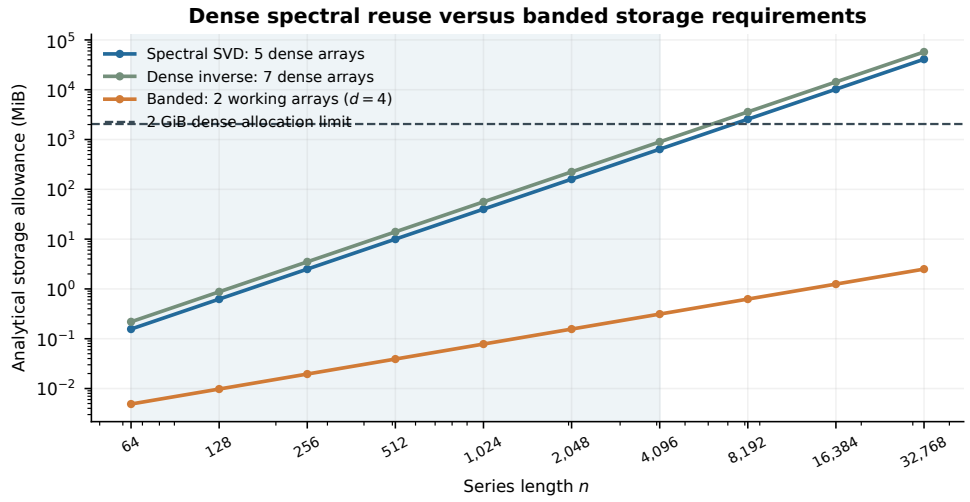


Fig. 9 Analytical source-code memory allowances as a function of series length for dense spectral SVD, an explicit inverse and two fourth-order banded working arrays. The shaded region denotes examined sample lengths and the dashed line the 2 GiB dense allocation policy. These curves are storage-policy calculations, not measured peak memory.

squares directly, and provides a common representation for multiple selection criteria. The advantage is pronounced in the matched CV/GCV grid comparisons and is compatible with the observed numerical-completion differences.

The recommendation should not be extended beyond the observed evidence. First, the banded method is often preferable when evaluating only one penalty. Second, the spectral basis and its work arrays require quadratic memory, which can dominate the decision for long series even when SVD is attractive at moderate n. Third, numerical acceptance is not interchangeable with certified forward accuracy: the two implementations apply different diagnostic checks, and both encounter smoothness-inversion difficulties in the larger fourth-order cases. Fourth, the recorded multi-process execution phases limit the transportability of wall-clock ratios to idle workstations or different parallel workloads.

These limitations suggest the following practical selection principle. If the task is a multi-candidate *complete-criterion* search and the dense spectral workspace satisfies a conservative memory budget, prepare the spectral decomposition once, reuse series-specific projections and precompute the candidate penalties when appropriate. If only one candidate is needed, or if dense allocation would be prohibitive, a structured banded method is a reasonable alternative subject to numerical residual checks. When a requested smoothness value cannot be inverted accurately, neither method should silently substitute an approximate grid winner: the affected comparison must be marked invalid or handled by a separately validated numerical strategy.

A further limitation is the restriction to synthetic data and one computational environment. Seed-cluster intervals quantify variability under the observed experimental protocol, but not variability across operating systems, processors, library versions or CPU utilization. The validity masks can induce survivor bias in timing ratios conditional on completed searches. The study does not establish improvements in trend-estimation risk or forecasting accuracy, and it does not constitute an independent high-precision verification of every evaluated penalty. Future work should isolate the source of the large-order smoothness-inversion failures, independently certify residual and criterion accuracy, and compare implementations under controlled processor occupancy.

6 Conclusions

This study establishes that the preferred computational method for penalized least-squares trend smoothing depends on the *parameter-selection workload*, not merely the cost of one linear solve. Across valid matched 100-candidate CV and GCV searches, spectral SVD consistently outperformed banded Cholesky in the representative tested configurations. A measured conditional-CV setting changed from favoring banded Cholesky for one candidate to favoring spectral SVD by a factor of approximately 4.85 for the complete grid. The numerical-status analysis further identified repeated banded post-solve failures on several complete grids, while retaining documented failure modes of the spectral pipeline.

We therefore recommend *spectral decomposition as the primary implementation for complete, repeated smoothing-parameter selection whenever its dense storage is affordable*. Banded methods remain valuable for isolated evaluations and long series, and numerical diagnostics remain mandatory for either implementation. The contribution is a measured, criterion-aware recommendation for computational practice rather than a new trend estimator or an unconditional claim of spectral superiority.

7 Code and data availability

The source implementations, analysis programs, selection-status records, derived timing data and reproducible figures are available in the public project repository at https://github.com/heri-espino/trendestimation. The underlying raw execution records are retained by the author but are not publicly deposited; their availability is subject to author approval. The publicly archived derived observations preserve incomplete and numerically inadmissible searches, and include the pairing and bootstrap metadata required to reproduce the tabulated comparisons.

Declarations

Funding. No specific grant support has been reported for this study.
Competing interests. No competing interests have been reported.
Ethics approval and consent. The study consists of synthetic numerical experiments and does not involve human participants, identifiable personal data or animal subjects.

Data and code availability. Analysis code, derived benchmark results, numerical-status tables and figures are available at https://github.com/heri-espino/trendestimation. Underlying raw timing files remain preserved by the author but are not part of the public repository; access is subject to author approval.

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